Yash Chaturvedi

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github.com/yashchat007 | yash-chaturvedi-portfolio.ai.studio | LeetCode | Codeforces

Education

Manipal University Jaipur — Jaipur, Rajasthan, India
Bachelor of Technology in Computer Science & Engineering; CGPA: 8.63/10 — *2025 – May 2029 (Expected)*

- **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Computer Networks

Experience

DRDO HQ, Ministry of Defence, Govt. of India — New Delhi, India
Software Engineering Intern – Security Analysis of High-Assurance Software — *May 2026 – Jul 2026*

- Evaluated a **Python**-based embedded file detection system that recursively identifies, extracts, classifies and summarizes concealed files across documents, images, audio, video and archives to mitigate **data exfiltration**
- Built an **adversarial validation suite** of security test payloads across **17 file formats**, simulating steganography, metadata manipulation, archive nesting, ZIP bombs, orphan payloads and invisible-character injection
- Ran cross-format validation to find detection blind spots and edge cases; proposed pipeline enhancements
- Audited Linux systems against security baselines with **OpenSCAP** and **Nessus**; reviewed PQC/DNSSEC research

Projects

FRIDAY – Local AI Desktop Agent | *Electron, React, TypeScript, Ollama, Playwright* — GitHub

- Built a local AI desktop agent that executes natural-language tasks through browser automation and Windows system tools
- Implemented **tool orchestration**, **approval-based actions**, task tracking, and a floating orb interface

RoleRadar – Internship Discovery Platform | *React, FastAPI, PostgreSQL, Playwright* — GitHub

- Built an automated internship discovery platform covering 8 companies, extracting job listings from official career websites and filtering opportunities by India eligibility
- Developed a resume-based matching engine with 4 weighted scoring dimensions—role relevance, skills, experience, and education—to rank internships and explain candidate-job fit

CineCircle – Social Movie Platform | *React, TypeScript, Spring Boot, PostgreSQL, Docker* — GitHub | Live Demo

- Built a full-stack app to discover movies via the **TMDB API**, manage watchlists and history, and rate films
- Secured the **Spring Boot** backend with **Spring Security** and **JWT**; added reviews, user follows, review interactions and **genre-based personalized recommendations**, containerized with **Docker**

Gesture Control – Hands-Free Desktop Control | *Python, OpenCV, MediaPipe, PyWin32, Tkinter, PyInstaller* — GitHub

- Built a Python desktop app for hands-free computer control via real-time webcam hand tracking with **MediaPipe**
- Implemented cursor control, pinch-to-click and drag-and-drop with personalized calibration and gesture recognition

Technical Skills

Languages: C++, Java, Python, JavaScript/TypeScript, SQL
Core: Data Structures & Algorithms, OOP, Low-Level Design, DBMS, Computer Networks
Tools: Git, Linux, PostgreSQL, Docker

Achievements & Certifications

- Featured in the **takeUforward (TUF) Hall of Fame** for DSA consistency and problem solving (220-day streak)
- **LeetCode Knight**: max rating **1863**, top **5.99%** globally; best contest rank 1633 of 30,000+
- **Codeforces Pupil**: max rating **1286**; solved **300+** problems, including 134 in a single month
- Solved **2000+ DSA problems** across LeetCode, Codeforces and takeUforward
- **Certifications:** Google Cybersecurity Certificate; Anthropic AI Fluency & Claude 101; EA Job Simulation (Forage)